2023

STATISTICS — HONOURS

Paper : CC-5

(Linear Algebra)

Full Marks : 50

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

1. Answer ***any five*** questions : 2×5

(a) What do you mean by linearly independent set of vectors?

(b) When can we say a vector $\underset{\sim}{\beta}$ to be a member of the vector space generated by the vectors $\{\underset{\sim}{\alpha}_1, \underset{\sim}{\alpha}_2, \ldots, \underset{\sim}{\alpha}_k\}$?

(c) Define a finite set of orthogonal vectors which are not orthonormal.

(d) Suppose A and B are two matrices of order $p \times q$, C and D are another two matrices of order $q \times r$. Using definition of matrix product, prove that

$$(A + B)(C + D) = AC + BC + AD + BD.$$

(e) Define the inverse of a square matrix of order m.

(f) How does rank of a matrix relate to its minors?

(g) Show that for every asymmetric matrix $B^{p\times p}$, we can always find a symmetric matrix $A^{p\times p}$ such that $\underset{\sim}{x}'B\underset{\sim}{x} = \underset{\sim}{x}'A\underset{\sim}{x}$, whatever $\underset{\sim}{x}^{p\times 1}$ may be.

(h) State any two properties related to eigen values and eigen vectors of a symmetric matrix.

2. Answer ***any two*** questions : 5×2

(a) If V_1 and V_2 are two vector subspaces, show that $V_1 \cap V_2$ is also vector subspace. Explain, with justification, what can be said about $V_1 \cup V_2$ in this regard.

(b) For any two matrices $A^{p\times p}$ and $B^{p\times q}$, show that $R(AB) \le R(B)$, equality attains when A is non-singular.

(c) Prove that all the diagonal elements of a positive definite (p.d.) matrix are strictly positive, and all its principal submatrices are also p.d.

3. Answer ***any three*** questions :

(a) (i) Show that any subset of a linearly independent set of vectors is always linearly independent. Explain, with justification, what you can say about subset of a linearly dependent set of vectors.

(ii) Prove that an orthogonal set of non-null vectors is necessarily linearly independent. Explain, with justification, whether a linearly dependent set of vectors can be orthogonal or not.

(3+2)+(3+2)

(b) (i) Show that basis of a vector subspace is not unique.

(ii) Suppose $\underset{\sim}{\beta}$ is a given vector belonging to a vector subspace V having a basis $\{\underset{\sim}{\alpha}_1, \underset{\sim}{\alpha}_2, ..., \underset{\sim}{\alpha}_k\}$. Prove that $\underset{\sim}{\beta}$ can be uniquely written as a linear combination of $\underset{\sim}{\alpha}_1, \underset{\sim}{\alpha}_2, ..., \underset{\sim}{\alpha}_k$.

(iii) What modification have to be done in the statement of (ii) above, if $\{\underset{\sim}{\alpha}_1, \underset{\sim}{\alpha}_2, ..., \underset{\sim}{\alpha}_k\}$ happens to be a spanning set instead of being a basis of V, and why? 3+3+(2+2)

(c) (i) What are elementary transformations and elementary matrices? Show that elementary transformations performed on the rows or columns of a matrix $A^{p \times q}$ is equivalent to pre-multiplying or post-multiplying A by some elementary matrix, respectively.

(ii) Show that every matrix of rank r can be written as the sum of r matrices each of rank unity. (3+4)+3

(d) (i) What is adjoint of a square matrix? If A is a square matrix of order p, show that $|Adj\ A| = |A|^p$. Also if $R(A) = p - 1$, find $R(Adj\ A)$. (2+2+2)+(2+2)

(ii) What is rank factorization? Show that rank of an idempotent matrix is its trace.

(e) (i) Stating all necessary results, briefly describe how do you use eigen values and eigen vectors to diagonalize a symmetric matrix.

(ii) Describe any other method of diagonalizing the matrix. 5+5